

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a deep blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene, digital aesthetic.

DL for ENG

LECTURE 10.1

Ionic Diffusion

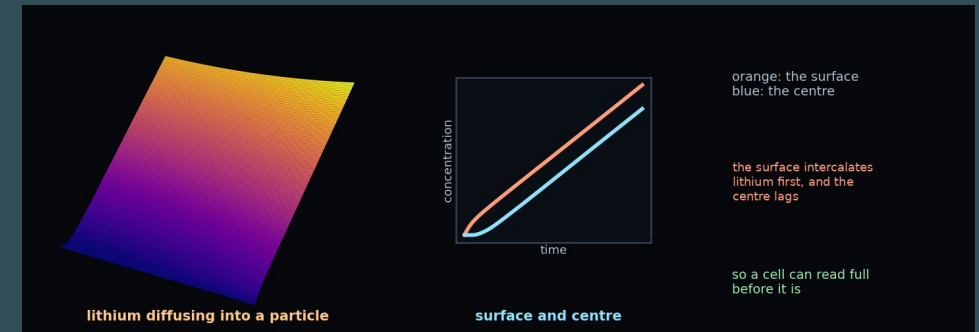
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Deep Learning for Engineering · 2026

PARTICLE · VOLTAGE · INVERSE



Lithium entering one particle: the surface intercalates lithium first and the centre lags, because lithium crosses only the surface and diffuses inward slowly. The lecture names the cell model's five fields - lithium in particles and electrolyte, two potentials, temperature - then solves the particle, reads the state of charge, and recovers the diffusivity from the surface.

Learning Objectives

SECTIONS, IN ORDER

- 1 **The cell** — five fields, two scales, and heat both ways
- 2 **The particle** — diffusion, scaling, state of charge
- 3 **PINN and inverse** — the start built in, the diffusivity found

Incropera carries the transport and the diffusion, Liu the inverse method; Gu & Wang (2000) the coupling, Chen et al. (2020) the LG M50 parameters. The example is Ex_10.1's graphite particle; L10.2 takes the same physics to a solid oxide cell.

READ, AND WHAT EACH ANSWERS

Incropera ch. 1 which transports share one law?

Incropera ch. 14 how does lithium reach the surface?

Liu ch. 4 what can readings identify?

Recap of L9.2 Gas Flow

1 - 2 · GAS AND SPECIES

1 Which three facts make the gas in a fuel channel a known, incompressible flow?

Its Mach number is below a thousandth, so the driving pressure drop is far too small to change the density. The electrode takes one H_2 and returns one H_2O , so the moles per volume and the velocity stay, though the mass density rises. At $\text{Re } 1.3$ the entrance is about a millimetre, so the parabola holds from the inlet. *White ch. 9; Ex_09.2 section 3*

2 Why can diffusion along the channel be dropped, and what does that make the equation?

The Péclet number along it is 125, so that term is a ten-thousandth of the one across. Without it the equation is parabolic in x : it marches from the inlet like heat in time and needs no outlet condition. A zero-slope outlet would be wrong, as the mean keeps rising. *Incropera ch. 14; Ex_09.2 sections 3 - 4*

3 - 4 · ONE PINN, EVERY SPEED

3 What does the PINN's trial function build in, and what does a loss term have to hold?

The hand calculation X/s , so the network learns only the profile across, in units of G . A loss term holds the network's flow-weighted mean at zero, or the balance drifts. The Sherwood number is a tenth of a per cent of the field: read wall and mean from the same column, and both methods find it to 0.3 %. *Incropera ch. 8; Ex_09.2 section 5*

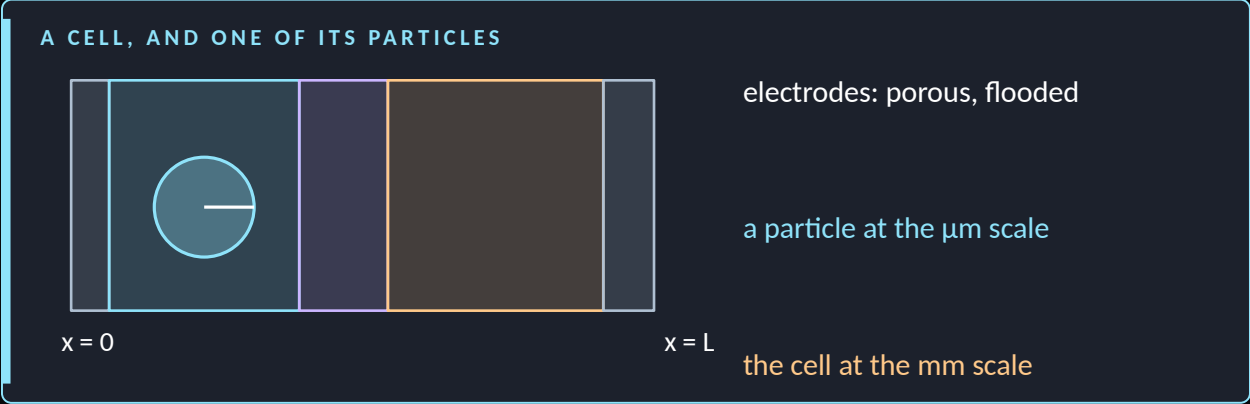
4 What can a PINN with the speed as an input do that a march cannot?

Give the field at any speed without solving again, and its derivative with respect to the speed by autograd. Newton's method then finds the speed for 25 % hydrogen at the outlet, 0.3174 m/s, the exact value, with no sweep. A new speed costs one evaluation; a march starts again. *Liu ch. 2.2; Ex_09.2 section 6*

Inside the Cell

Lithium-ion cell — a sandwich of two porous electrodes around a separator, flooded with electrolyte. Lithium lives in two places at once: inside the solid particles, and dissolved in the liquid between them. A model of it solves five fields on two length scales.

THE SANDWICH, AND THE SCALE INSIDE IT



The cell is millimetres across; a particle is micrometres. Lithium must diffuse from each particle's interior to its surface before it can leave, so the model has a coordinate inside a coordinate.

THE FIVE FIELDS OF A CELL MODEL

field	lives	equation
c_s in the particles	0 to R	parabolic
c_e in the electrolyte	0 to L	parabolic
ϕ_s in the solid	0 to L	elliptic
ϕ_e in the electrolyte	0 to L	elliptic
T , the temperature	the cell	parabolic

Every type in the table is one this course has solved: L8's heat equation and its Poisson problems, wearing electrochemical names. What is new is that the five supply terms to each other. Ex_10.1 solves the first row; the rest is this lecture's.

Two Phases Two Potentials

Charge conservation — two elliptic equations, one per phase, with no time derivative: at every instant the potentials adjust to carry the current. The reaction current appears in both with opposite signs, because charge that leaves the solid enters the electrolyte.

THE TWO EQUATIONS, AND THEIR SHARED SOURCE

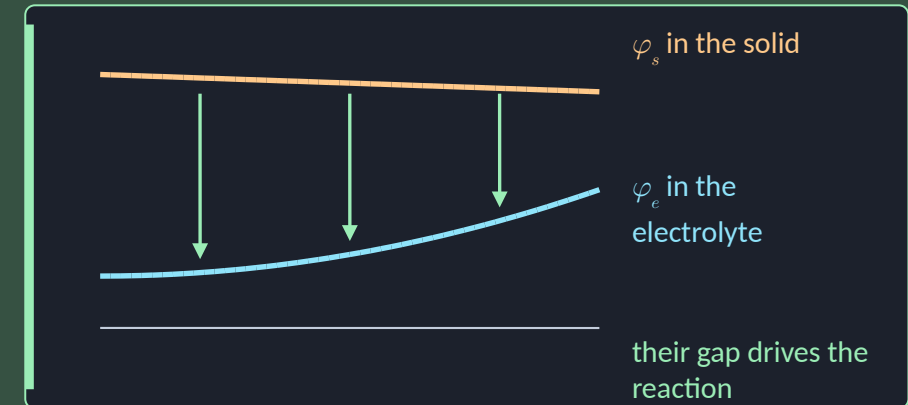
$$\nabla \cdot (\kappa^{eff} \nabla \phi_e) + \nabla \cdot (\kappa_D^{eff} \nabla \ln c_e) + j^{Li} = 0$$

$$\nabla \cdot (\sigma^{eff} \nabla \phi_s) - j^{Li} = 0$$

ϕ_e, ϕ_s the potentials of electrolyte and solid, $\kappa^{eff}, \sigma^{eff}$ their effective conductivities
 c_e the lithium in the electrolyte, j^{Li} the reaction current per volume, opposite signs

TALK • 2 MIN How would you explain, in your own words, what an equation with no time derivative says about its field? Then, in the drawing on the right, what happens to the green arrows where the two curves run parallel?

ACROSS ONE ELECTRODE



the extra term a gradient of c_e drives ϕ_e

why the two ions move unequally

The gap $\phi_s - \phi_e$ drives the reaction.

Three Model Levels

Model hierarchy — the three levels of a cell model: the single particle model (SPM) keeps one particle per electrode, the SPMe adds the electrolyte, the DFN keeps a particle at every point of the electrode. The level is chosen by the current, not by ambition.

PICK THE LEVEL BY THE CURRENT

model	solves	holds to
SPM	two particles, one current	about 1C
SPMe	adds c_e and ϕ_e along x	about 3C
DFN	a particle at every node of x	any rate

For a PINN the choice costs more than for a classical solver: each added field is another network, another residual, another loss term to balance. A matrix only grows.

WHAT THE EXERCISE KEEPS

the field	one particle's $c_s(r, t)$
the current	shared evenly: j known
the voltage	algebra after the solve
left out	electrolyte, potentials, heat

In the single particle model only the diffusion is a differential equation. Charge conservation is trivial - identical particles share the current evenly - and the voltage is one line of algebra once the particle is solved. MP10.1A takes the full DFN on.

Heat Runs Both Ways

Thermal coupling — the loop between electrochemistry and temperature: the reaction, its entropy and the resistances make heat, and the temperature changes the diffusivities and rates that make it. A model with properties held at one temperature gets the trend backwards.

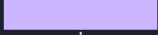
THE ENERGY EQUATION, AND ITS THREE SOURCES

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q, \quad q = q_{rxn} + q_{rev} + q_{ohm}$$

ρc_p the heat capacity per volume, k the thermal conductivity, q_{rxn} reaction, q_{rev} entropic, q_{ohm} ohmic

THE THREE SOURCES

reaction q_{rxn}  two terms only heat

entropic q_{rev} 

ohmic q_{ohm} 

cools heats

one changes sign with the current

WHAT GU & WANG FOUND

coupled model warming helps: transport improves

decoupled model sees the heating, not the help

the result the trend backwards

in a large cell tens of degrees across it

for a PINN one more residual term

The entropic coefficient was long unmeasured; O'Regan et al. (2022) supply it for the LG M50.

Diffusion Inside a Particle

Solid diffusion — Fick's second law in a sphere: L8.2's heat equation in spherical coordinates, with symmetry at the centre and the reaction's flux at the surface. The reaction sees the surface concentration, not the average, so a particle can read as empty before it is.

THE EQUATION, AND ITS TWO CONDITIONS

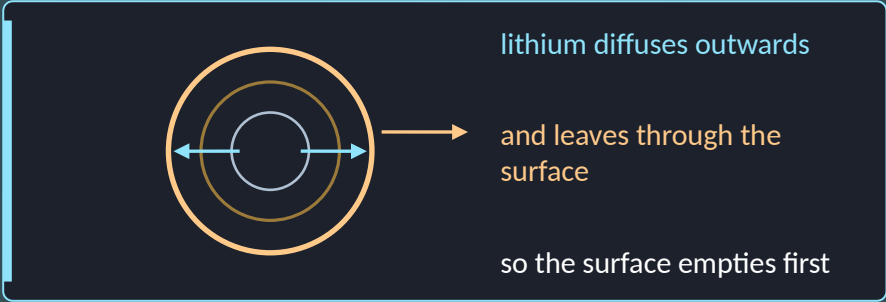
$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_s \frac{\partial c_s}{\partial r} \right)$$

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0, \qquad -D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R} = j$$

c_s the lithium in the particle, r the radius, D_s the solid diffusivity, R the particle radius, j the surface flux

TALK • 2 MIN How would you explain, in your own words, why the reaction sees the surface and not the average? Then, in the drawing on the right, what happens to the gap between surface and centre when the current doubles?

EX_10.1'S GRAPHITE PARTICLE



	value
radius R	5.86 μm
diffusivity D_s	$3.3 \times 10^{-14} \text{ m}^2/\text{s}$
diffusion time R^2/D_s	1041 s
discharge	1C: 5 A for an hour
<i>Chen et al. (2020), as PyBaMM distributes them. The hour and the diffusion time are alike: the profile matters.</i>	

Scaling the Problem

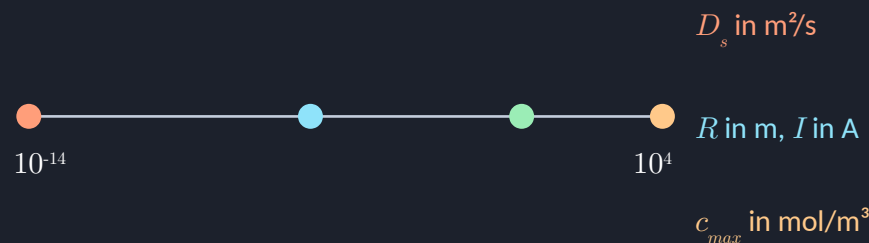
Scaling — measuring each quantity in a unit the problem itself supplies, so every number the network meets is of order one. In raw SI units these quantities span about twenty orders of magnitude, one residual term dwarfs the rest, and no loss weight makes up for that.

THE SCALED PROBLEM, AND ITS ONE CONSTANT

$$u_\tau = C(u_{rr} + \frac{2}{r}u_r), \quad C u_r = 1 \text{ at } r = 1, \quad C = \frac{D_s t_{end}}{R^2} = 3.46$$

u the lithium lost, over $c_{ref} = j t_{end}/R$, r over R , τ over the hour t_{end} , C the scaled diffusivity

THE RAW QUANTITIES, ONE LOG SCALE



THE SCALES, SUPPLIED BY THE PROBLEM

length	the radius R
time	the hour t_{end}
concentration	$c_{ref} = j t_{end}/R$
what is left	one number, $C = 3.46$

One free gift of these units: the mean of u over the sphere is exactly 3τ - what has left is the current times the time - whatever the diffusivity. The mass balance, and section 5 puts it in the loss.

State of Charge and Voltage

State of charge — how full the cell is. The bulk value is the mean concentration: a coulomb counter, exactly $1 - \tau$ here, needing no field. A gauge that has lost count must read the voltage, and the voltage sees only the surface - it is the solved field that converts the one into the other.

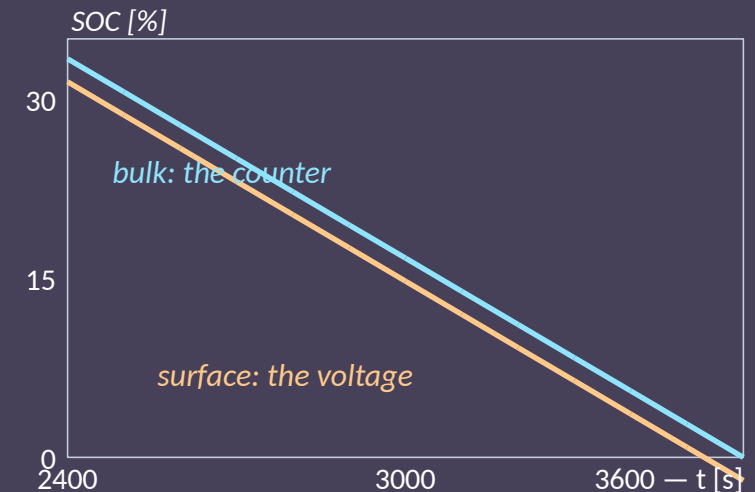
THE SURFACE SETS THE VOLTAGE

$$\theta = \frac{c_{se}}{c_{s, \max}} \quad V = U_p(\theta_p) - U_n(\theta_n) + \eta_p - \eta_n$$

θ the surface state of charge, c_{se} the surface concentration, $c_{s, \max}$ the most a particle holds U_p, U_n the measured open-circuit curves, η_p, η_n the overpotentials (left out in Ex_10.1)

The open-circuit curves are measured, not derived: strongly nonlinear, with plateaus where two phases coexist. On LFP the plateau moves some 10 mV over most of the charge, so there the voltage says almost nothing about the surface - Ex_10.1's film shows it.

THE TWO GAUGES, OVER THE HOUR



Ex_10.1's exact field, the last 20 minutes: the surface reads about 2 points low and crosses zero a minute early - the cell cuts off with charge still in the bulk.

A PINN for the Particle

Trial function with the start built in — a network wrapped so the initial condition holds whatever the weights do: τ times the network is zero at the start, as L8.2's pulse was. The residual is multiplied by r , so it has no $1/r$ in it and at the centre it reduces to the symmetry condition for free.

THE TRIAL FUNCTION, AND THE BALANCE IN THE LOSS

$$\hat{u}(r, \tau) = S \tau \mathcal{N}(r, \tau), \quad \bar{\hat{u}} = 3\tau$$

$S = 3$ the size of the answer, $\tau \mathcal{N}$ zero at the start whatever the network does, $\bar{\hat{u}}$ its mean over the sphere

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=2, n_hidden=32, n_layers=4) # r and tau in
u = 3.0 * tau * net(p) # the trial function: zero at the
start
g = grad(u, p) # first derivatives by autograd
res = r * g[:,1:2] - C * (r * d2(u, p, 0) + 2 * g[:,0:1]) # the residual, times
mass = (u(q_pts) * q_w).sum(1) - 3 * tau_q # the mass balance, held at 0, w
= 1000
```

THREE METHODS, ONE TRUTH

	FVM	PINN
surface at 3600 s	890 mol/m ³	899 mol/m ³
worst error	29 mol/m ³	21 mol/m ³
the hour, online	a few ms	under 1 ms
training, offline	none	about 45 s
agreed 30 mol/m ³	40 shells	met

The exact series is the truth: surface 891, centre 2260 mol/m³ at the end. The mass balance must be a loss term: a network a little late with the flux at the start loses lithium it never gets back. The surface points exclude $\tau = 0$, where the trial function has no gradient.

The Diffusivity From the Surface

Identifiability — whether the data can tell the parameter at all. The voltage reads the surface through the open-circuit curve, and the surface's lag behind the mean is what carries the diffusivity. On LFP's plateau that reading goes nearly blind.

ONE MORE TRAINABLE NUMBER

$$C \text{ trainable, } \mathcal{L} = \mathcal{L}_{PDE} + w_S \mathcal{L}_{flux} + w_M \mathcal{L}_{mass} + w_D \mathcal{L}_{data}$$

$\mathcal{L}_{data}$ the misfit to 37 surface readings, $w_S = 10$, $w_M = 1000$ hold the flux and the mass balance

THE INVERSE PROBLEM IN PYTORCH

```
logC = torch.nn.Parameter(torch.log(C0))    # the diffusivity, trained with
the weights
C = torch.exp(logC)                        # positive by construction
L = res(C).pow(2).mean() + W_D * (u_surf - readings).pow(2).mean() # physics +
data
data.backward(); opt.step()                # one step moves weights and C
together
```

WHAT EX_10.1 RECOVERS

the data 37 surface readings

the noise 50 mol/m³ on each

the start half the true value

found D_s 0.38 % low, in a minute

not divided by C divided, C runs away

Unlike Ex_08.2: divided by C , a uniform particle fits the readings and C grows without limit.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the electrolyte, the SPM	lithium and its potential between the particles	Incropera ch. 14	above about 1C
<u>Butler–Volmer kinetics</u>	the exponential rate at the interface	PyBaMM docs	the overpotentials in the voltage
<u>the full model, the DFN</u>	a particle at every point of the electrode	Marquis et al. (2019)	above about 3C; MP10.1A
heat in two dimensions	the core runs hotter than the can	Incropera ch. 5	large cells and packs
<u>ageing and lost lithium</u>	slower diffusion, capacity gone for good	O'Regan et al. (2022)	the BMS; MP10.1B

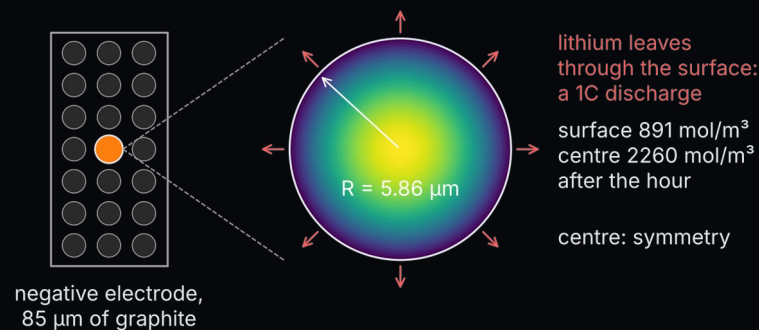
WHY THEY ARE LEFT OUT

Ex_10.1 solves one particle with a known flux, scored on an exact series, so none of them is needed to do it. In the single particle model the current is shared evenly and the voltage is algebra; the rows return in the two mini projects.

Exercise 10.1

ONE NOTEBOOK, SECTION BY SECTION

- 1 - 3 — the particle, its data, its scales, the mass balance
- 4 — the exact series; finite volumes; the gauges and the film
- 5 — a PINN with the start built in; the three compared
- 6 — the diffusivity from 37 noisy surface readings
- 7 - 9 — the report; mini projects MP10.1A and MP10.1B



THE QUESTION EACH PART ANSWERS

- 1 - 3 which scales make it order one?
- 4 what does the surface hide?
- 5 can the PINN match the grid?
- 6 what do readings identify?

REPORT QUESTION, TO CONCLUDE

The grid solved the particle in milliseconds; the network matched it and then recovered the diffusivity from noisy readings in the same run. For which questions about a battery would you train a network, and for which solve a grid?

Key Takeaways

THE CELL MODEL

what it is	five fields on two length scales
function	the voltage from the fields
train, tune	pick the level by the current
vs one physics	each field feeds the others
used for	what a BMS cannot measure

HEAT AND THE INVERSE

what it is	a particle PINN, start built in
function	the field, and D_s from readings
train, tune	scaled units; balance in the loss
vs finite volumes	equal accuracy, plus the inverse
used for	the parameter that ages most

Next — L10.2: the same transport and charge, in a ceramic cell at 800 °C.

Questions

1 – 2 • A NETWORK FOR THE PARTICLE

- 1** How can a network learn the lithium inside a particle, and what should it not have to learn?

It takes position and time and is trained until the diffusion equation holds everywhere, with no solved data. What is known for certain, the starting state and the symmetry about the centre, is built into its form, so it spends its training only on what is unknown. *Liu ch. 2; Ex_10.1 section 5*

- 2** Why does a network need the particle problem in the problem's own units?

In everyday units the quantities differ by many orders of magnitude, and the network's training sees only the largest of them. Measured against the particle's own size, time and current, every quantity is about one, so each part of the physics is learned with equal care. *Ex_10.1 section 3*

3 – 4 • CONSERVATION AND THE INVERSE

- 3** Why is the conservation of lithium not automatic for a network, and how is it kept?

A network satisfies the equation only approximately, and small errors at the surface add up to lithium created or lost. The total that has entered is known from the current, so training is told to hold it, and the field stays physically possible. *Liu ch. 2; Ex_10.1 section 5*

- 4** How can a network reveal a property inside a particle from outside measurements, and when can it not?

Leave the unknown property free and train it with the network, so the field both obeys the physics and matches what is measured. It works when the measurements change with that property; where they barely move, as on LFP's flat voltage, nothing can recover it. *Liu ch. 4; Ex_10.1 section 6*

NEXT

L10.2 — Solid Oxide Cells

The same transport and charge conservation in a ceramic cell at 800 °C, where L9.2's steam channel meets its electrode.

BEFORE THE NEXT SESSION

read	Incropera ch. 14
run	Ex_10.1, sections 1 to 6
bring	your diffusivity, and its error
think about	what a BMS can and cannot see

Play the film at 1C and at 5C before the session, and watch what the voltage does on the LFP plateau.